

University of Debrecen, Faculty of Informatics

Logic in computer science



Zero-order logic

(propositional logic)

normal forms

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Literal

Let $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language.

If $p \in Con$ then

- the p and $\neg p$ formulas are called *literals*, where
- the p atomic formula is the *base of the literal*;
- the p literal is a *positive literal*,
- the $\neg p$ literal is a *negative literal*.

Elementary conjunction, elementary disjunction

An *elementary disjunction*:

- is a literal, or
- a disjunction of a literal and an elementary disjunction.

$$\ell_1 \vee \ell_2 \vee \cdots \vee \ell_n \quad \text{where } n \geq 1 \text{ and } \ell_i \text{ are literals for all } i = 1, \dots, n$$

An *elementary conjunction*:

- is a literal, or
- a conjunction of a literal and an elementary conjunction.

$$\ell_1 \wedge \ell_2 \wedge \cdots \wedge \ell_n \quad \text{where } n \geq 1 \text{ and } \ell_i \text{ are literals for all } i = 1, \dots, n$$

Normal forms

A *disjunctive normal form* (DNF):

- is an elementary conjunction, or
- a disjunction of an elementary conjunction and a disjunctive normal form.

$e_1 \vee e_2 \vee \cdots \vee e_n$ where $n \geq 1$ and e_i is an elementary conjunction for all $i = 1, \dots, n$

A *conjunctive normal form* (CNF):

- is an elementary disjunction, or
- a conjunction of an elementary disjunction and a conjunctive normal form.

$e_1 \wedge e_2 \wedge \cdots \wedge e_n$ where $n \geq 1$ and e_i is an elementary disjunction for all $i = 1, \dots, n$

Normal form theorem

Let $\mathcal{L}^{(0)} = \langle LC, Con, Form \rangle$ be a zero-order language and $A \in Form$ be a formula.

1. There exists a formula $D \in Form$ such that $A \Leftrightarrow D$ and D is a disjunctive normal form.
2. There exists a formula $C \in Form$ such that $A \Leftrightarrow C$ and C is a conjunctive normal form.

The *disjunctive normal form* of a formula A is a formula in disjunctive normal form which is semantically equivalent with A .

The *conjunctive normal form* of a formula A is a formula in conjunctive normal form which is semantically equivalent with A .